

HOME AUTOMATION SYSTEM USING VOICE COMMANDS

Revolutionizing Home Control Through
Technology

INTRODUCTION:

- What is home automation?**

Home automation means controlling your home devices (like lights, fans, or heater) with technology, often using your phone or voice commands.

- Why is home automation useful?**

It makes life easier, saves time, and helps save energy by controlling devices automatically.

- Voice control in home automation**

You can give voice commands (like “turn off the lights”) to control smart devices in your home.



Designed By : Subin Surendran Archt
Panampilly Nagar, Kochi, Kerala. Ph: +91 9
Email:

medesign.blogspot.com

AGENDA

OUR TASKS INCLUDE:

- To turn on and turn off the light
- To turn off water heater(bulb taken as load) after certain minutes
- To control the light brightness
- To change the colors of light according to PWM values of RGB

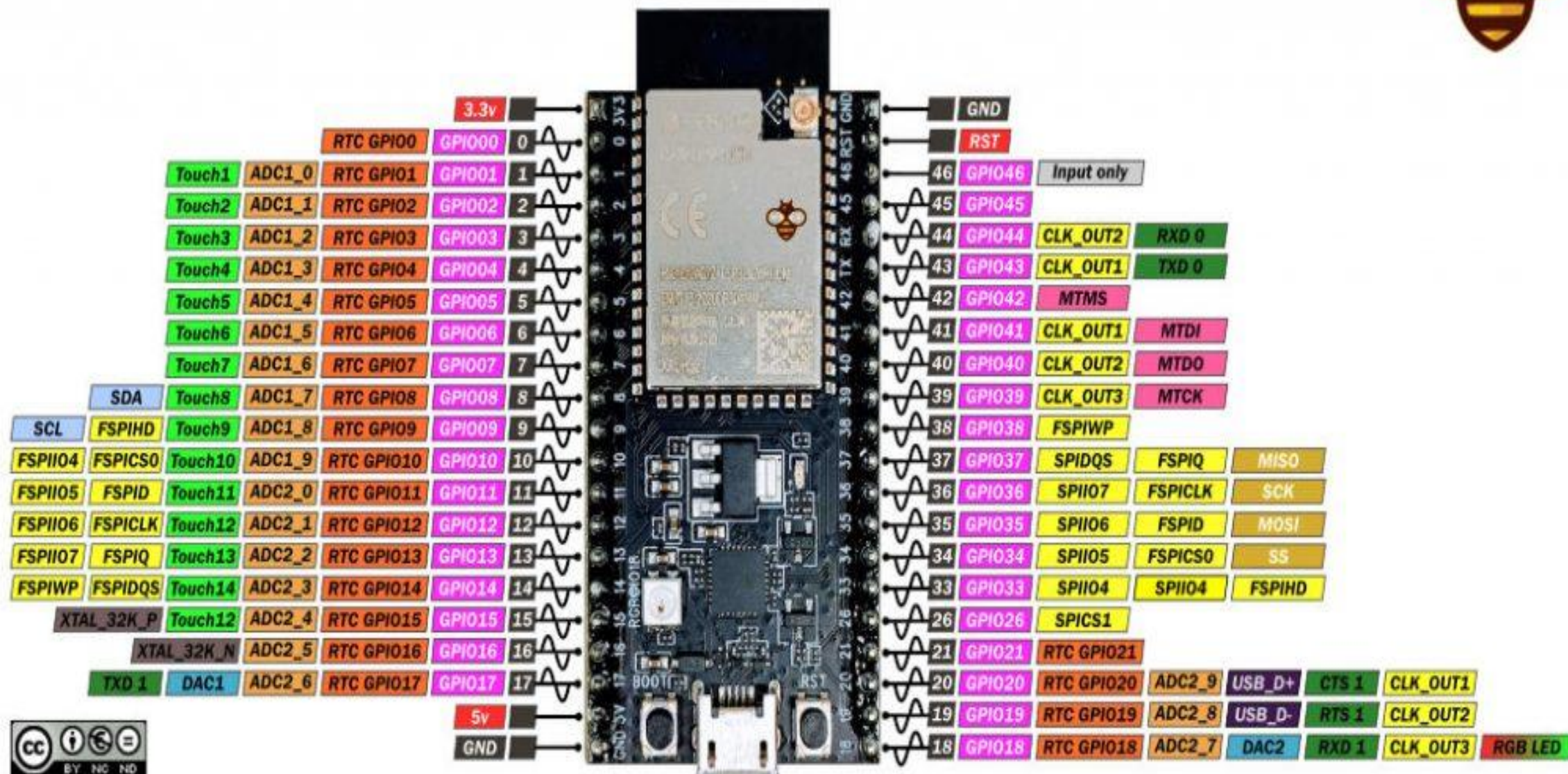
COMPONENTS

Hardware Components:

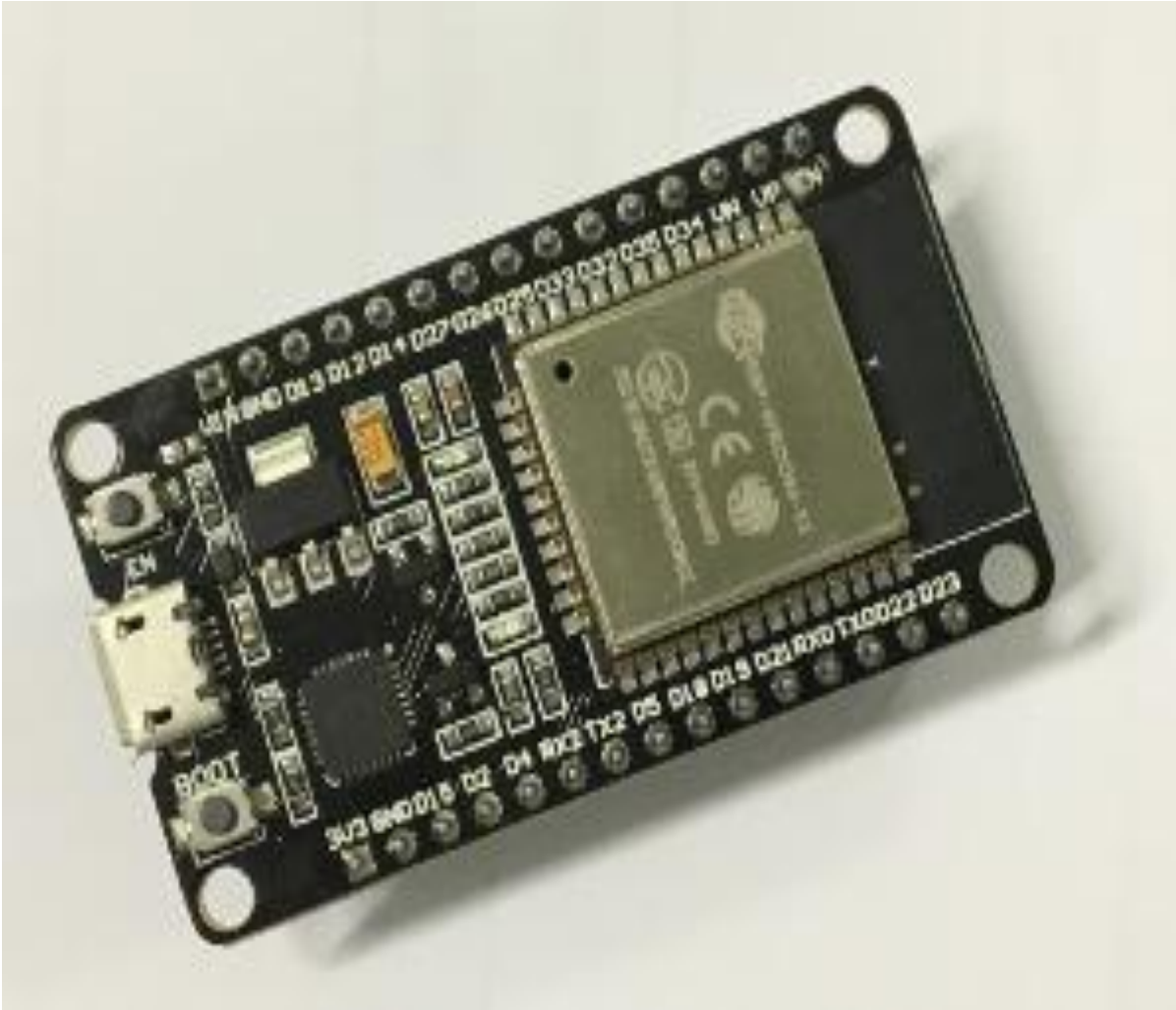
- Relay Module
- LED bulb
- RGB LED
- Power Supply
- ESP-32

Software Components

- Arduino IOT cloud
- Google Assistant
- Thonny ide



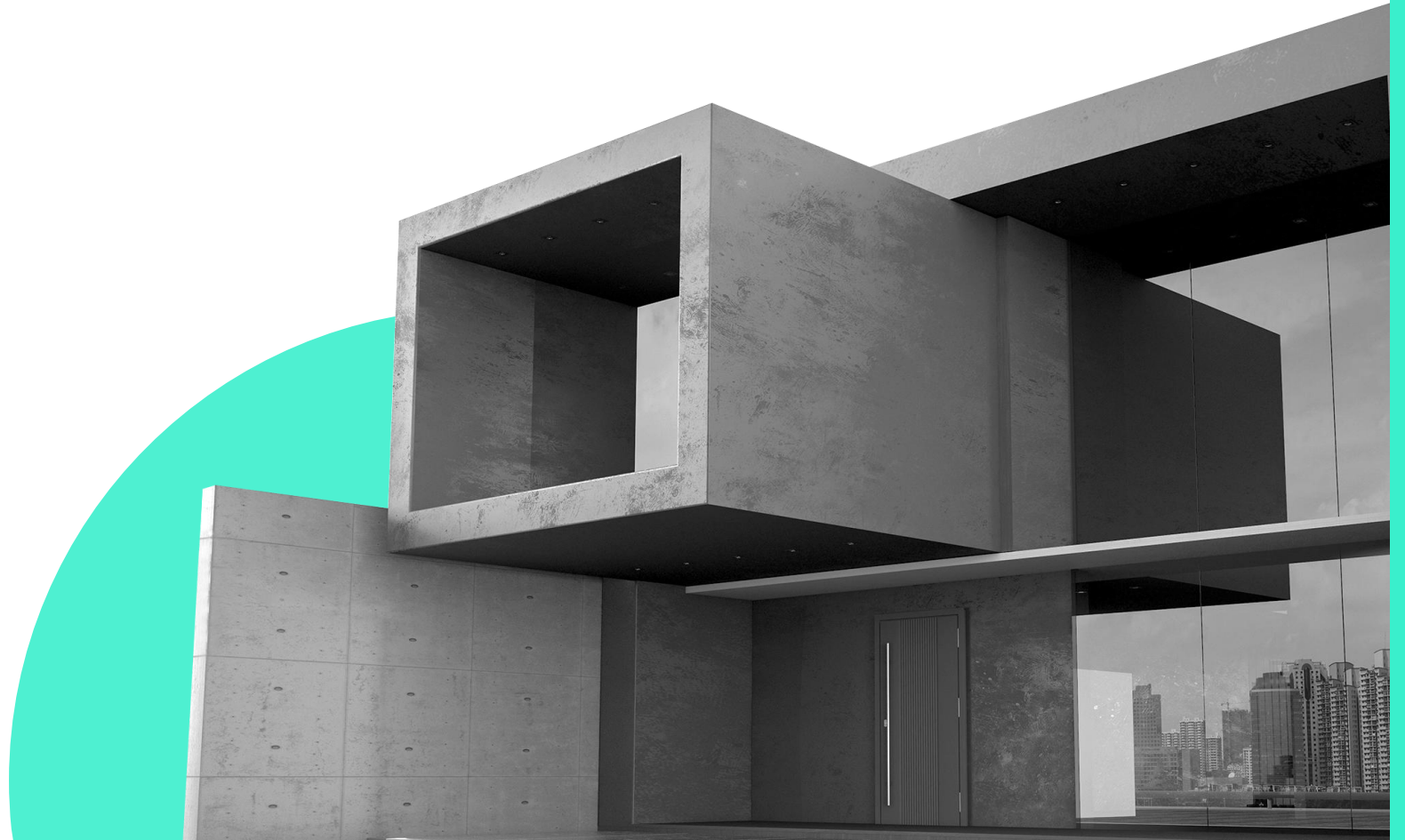
Features of ESP32:



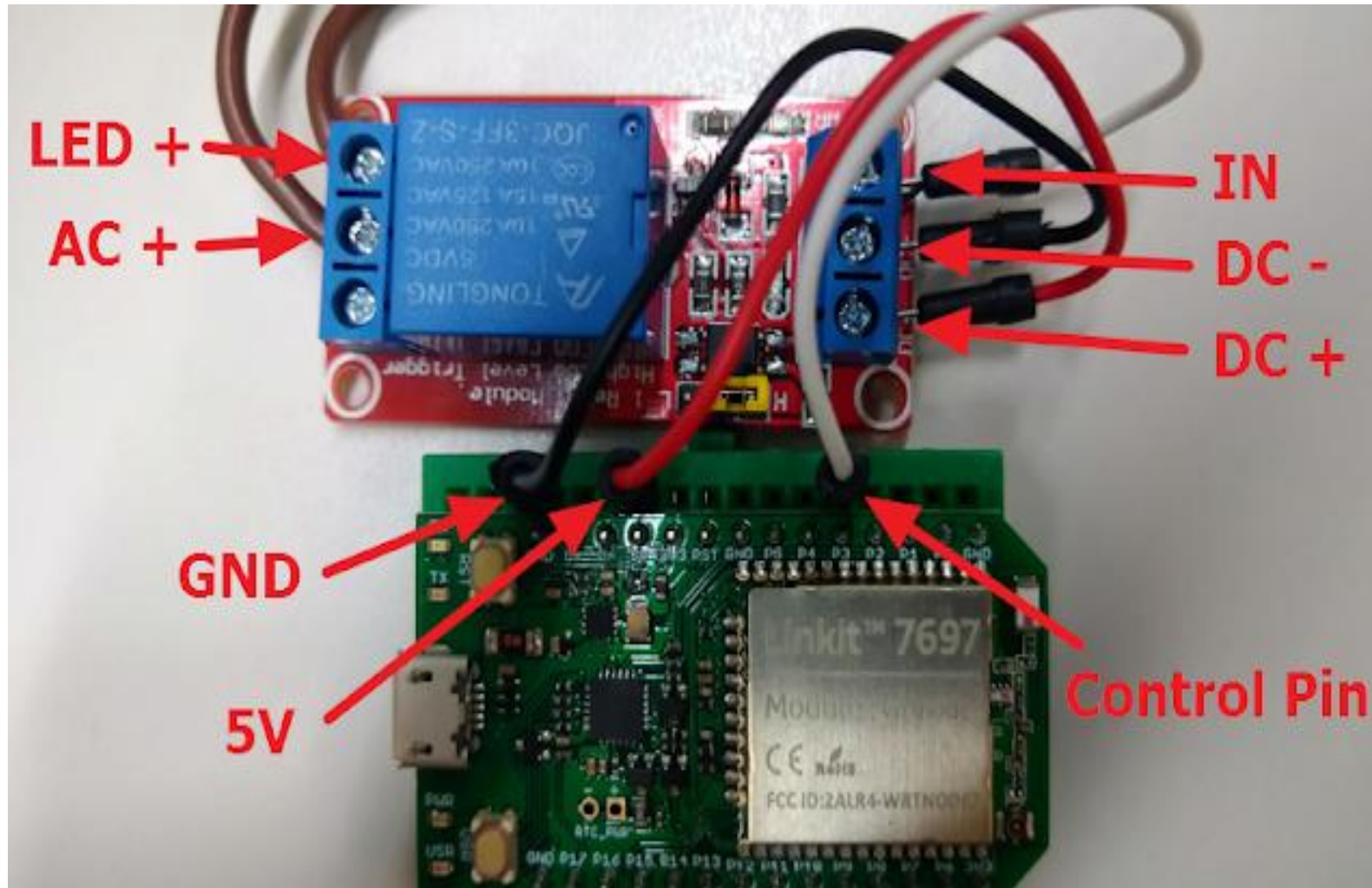
- **Dual-core 32-bit processor** (up to 240 MHz)
- **Wi-Fi + Bluetooth (Classic & BLE)**
- **520 KB SRAM** + support for external flash/PSRAM
- **Multiple GPIOs** with ADC, DAC, PWM, Touch, etc.
- **SPI, I2C, UART, CAN, I2S** support
- **Low power modes** for battery use
- **Secure boot + Flash encryption**
- **Supports Arduino, MicroPython, ESP-IDF**

LIBRARIES USED:

- machine
- time
- network
- logging
- arduino_esp8266_cloud
- Corb2
- Senml



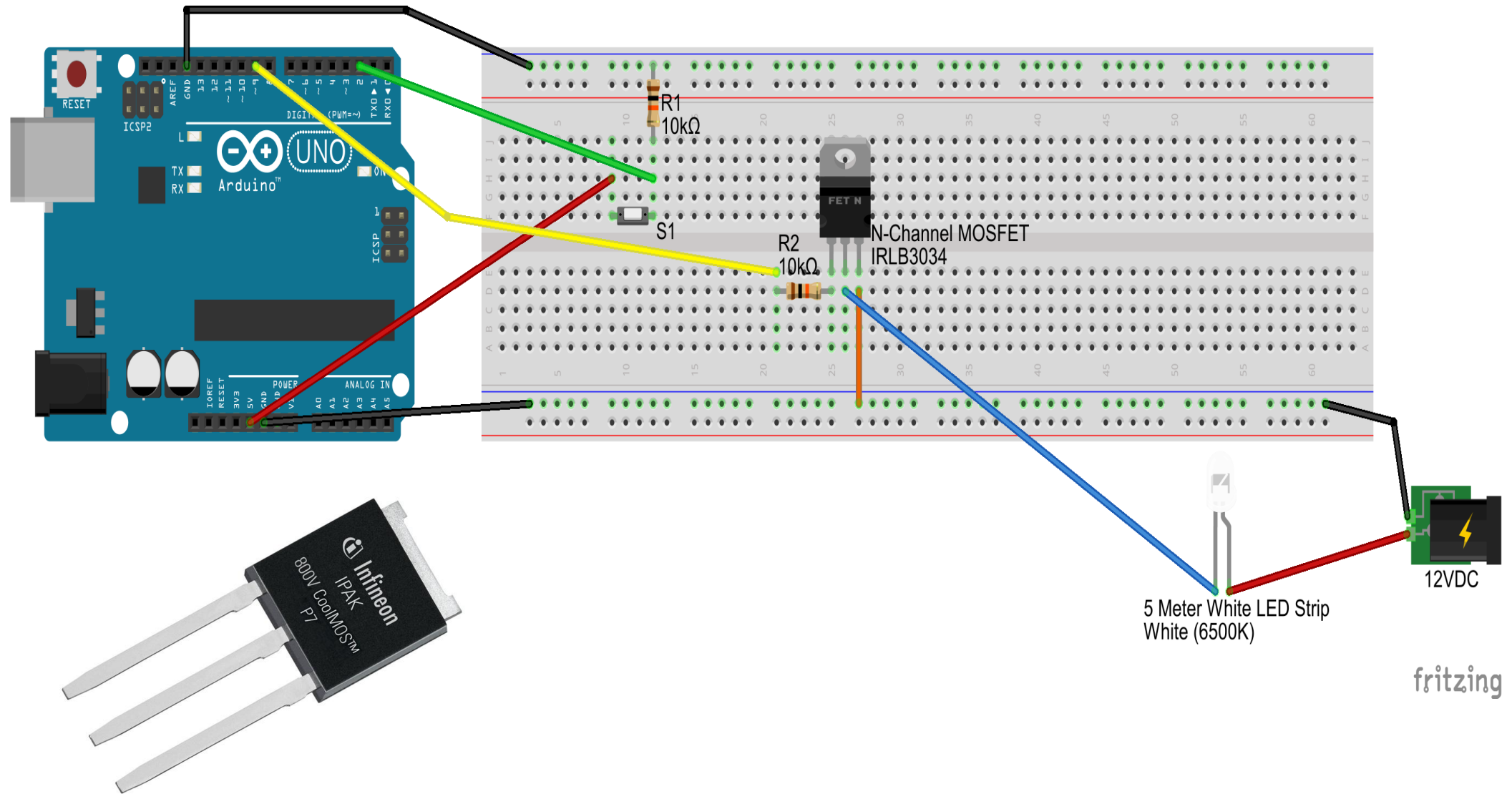
RELAY MODULE



TURN ON AND TURN OFF THE LIGHT:

- The voice command is sent to the Arduino IoT Cloud via Google Assistant or Alexa.
- The microcontroller receives the signal from the cloud.
- GPIO pin is triggered to turn the light ON or OFF using a relay or output pin.
- Useful for daily use and helps in energy saving.





fritzing

TURN OFF WATER HEATER (LOAD BULB)

- When the light is turned ON, a timer starts.
- After the set time (default: 15 minutes), the light turns OFF automatically.
- This is done using a Timer in Micro Python
- The time can be changed through the Arduino IOT cloud
 - 👉 For example, from 15 mins to 5 mins.
- Useful for controlling a water heater or any device that should turn off by itself.

**INCREASES
SAFETY**

**HANDS-FREE
OPERATION**

**SMART
SCHEDULING**

CONTROLLING LED LIGHT BRIGHTNESS

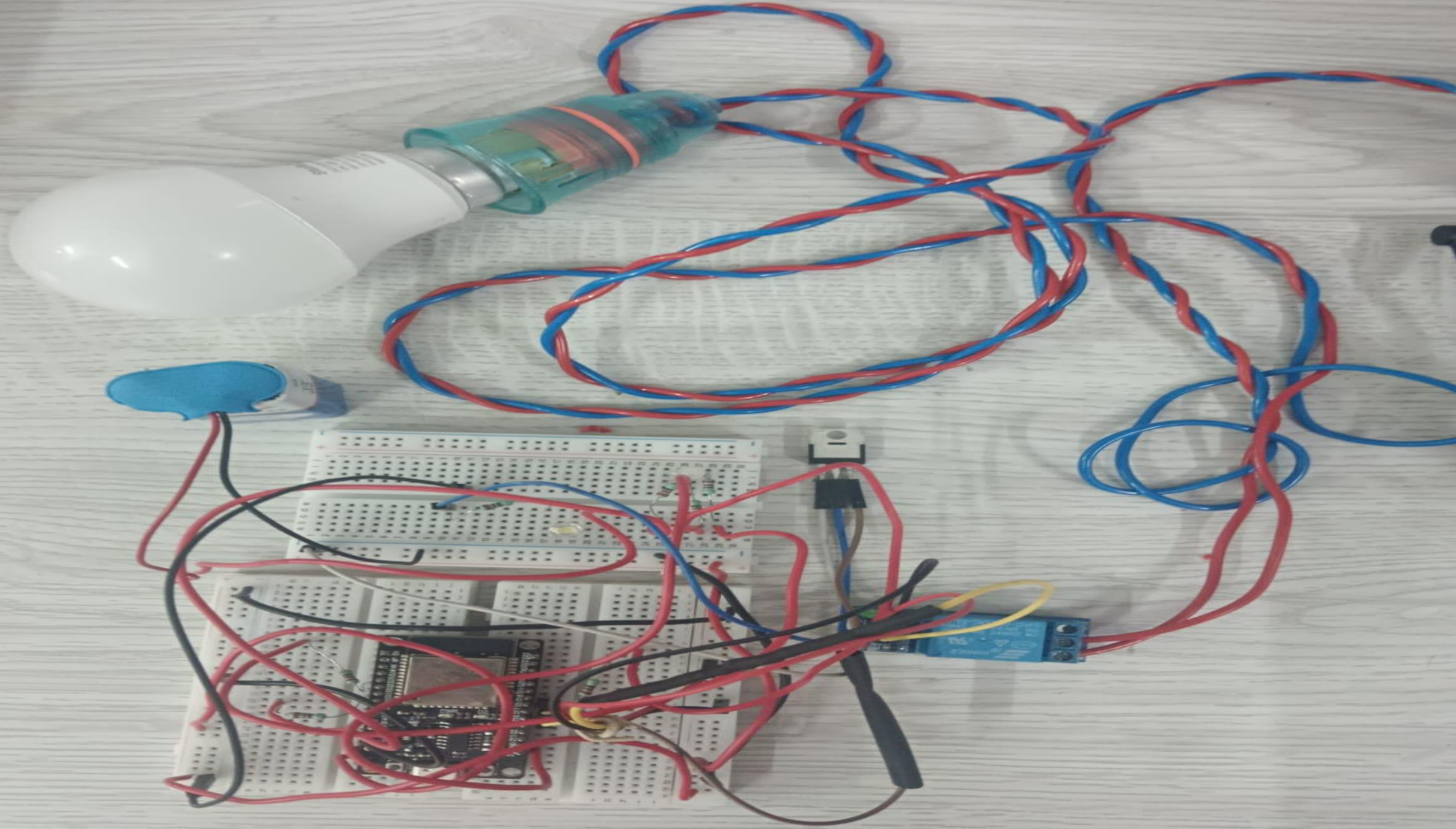
- Brightness of **Light 2** is controlled using **PWM (Pulse Width Modulation)**.
- A value (0 to 100%) is sent from the cloud through voice
- This value is converted to PWM duty cycle in code:
👉 `duty = int((value / 100) * 1023)`
- The duty value is applied to the PWM pin:
👉 `light2_pwm.duty(duty)`
- Voice Command Example:
👉 “Set brightness to 50%”



**ENERGY
SAVING**

**HANDS-FREE
OPERATION**

**EXTEND BULB
LIFE**



CHANGE COLOR OF THE LIGHT BULB

- Light 3 is an **RGB LED** (Red, Green, Blue).
- The color is set using **voice commands**
- Each color (R, G, B) has a value between **0 and 255**.
- These values are converted into PWM duty cycles:

👉 **Example:**

```
r = int((value.get("r", 0) / 255) * 1023)
```

- PWM is applied to control LED brightness for each color:

👉 `red_pwm.duty(r), green_pwm.duty(g), blue_pwm.duty(b)`

- This allows **millions of color combinations**.

**MOOD
LIGHTING**

**BETTER EYE
COMFORT**

**PERSONALIZED
EXPERIENCE**

BENEFITS OF THIS PROJECT:

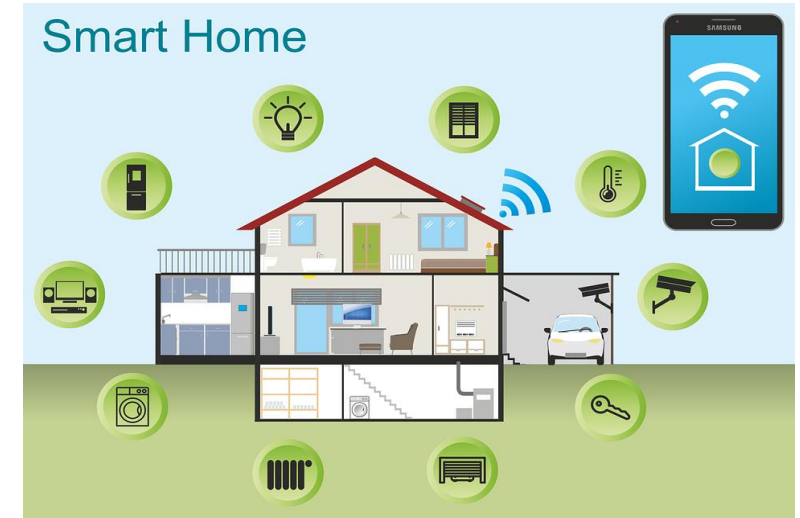
- **Convenient & Hands-Free** — Control your home with just your voice
- ☒ **Energy Efficient** — Devices turn off automatically, brightness is adjustable
- ☒ **Safety Improved** — Auto shutoff prevents overheating
- ☒ **User-Friendly** — Easy setup with smartphone and cloud
- ☒ **Customizable** — Adjust brightness, colors, and timers as needed
- ☒ **Scalable** — More devices and sensors can be added in the future
- ☒ **Modern Living** — Smart homes enhance lifestyle and comfort



[This Photo](#) by Unknown Author is licensed under [CC BY-NC-ND](#)

CONCLUSION:

- This project shows how home automation using voice commands can make everyday life easier, safer, and smarter.
- With features like auto shut-off, brightness and color control, and cloud connectivity, users gain full control over their home environment.
- It promotes energy saving, supports modern smart living, and can be upgraded with more devices.
- In short:
"A step forward to building a smarter, more comfortable home."





Thank
you

SUBMITTED BY:
AP23110011601
AP23110011594
AP23110011661
AP23110011647